Equilibrium Calculations When K_c Is Very Small

Many chemical reactions have equilibrium constants that are either very large or very small. For example, most weak acids have very small values for K_c.

Therefore, only very tiny amounts of products form when these weak acids react with water.

When the K_c for a reaction is very small the equilibrium calculations can usually be simplified considerably.

EX.16

Hydrogen, a potential fuel, is found in great abundance in water. Before the hydrogen can be used as a fuel, however, it must be separated from the oxygen; the water must be split into H_2 and O_2. One possibility is thermal decomposition, but this requires very high temperatures. Even at 1000 °C, $K_c = 7.3 \times 10^{-18}$ for the reaction:

$$2H_2O(g) = 2H_2(g) + O_2(g)$$

If at 1000 °C the H_2O concentration in a reaction vessel is set initially at 0.100 M, what will the H_2 concentration be when the reaction reaches equilibrium?

Solution

We know that at equilibrium:

$$\frac{[H_2]^2[O_2]}{[H_2O]^2} = 7.3 \times 10^{-18}$$

We now set up the concentration table.

Initial Concentrations

The initial concentration of H_2O is 0.100 M; those of H_2 and O_2 are both 0.0 M.

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Changes in Concentration We know the changes must be in the same ratio coeffi -cients in the balance d equation, so we place x's in this row with coefficients equal to those in the chemical equation. Because there are no products initially, their changes must be positive and the change for the water must be negative. This gives

	$2H_2O(g)$	$= 2H_2(g)$ +	$O_2(g)$
Initial concentration(M)	**0.1**	**0.0**	**0.0**
Change in concentration(M)	**-2x**	**+2x**	**+x**
Equilibrium concentration(M)	**0.1-2x**	**2x**	**x**

When we substitute the equilibrium quantities into the mass action expression we get

$$\frac{(2x)^2 x}{(0.100 - 2x)^2} = 7.3 \times 10^{-18}$$

or

$$\frac{4x^3}{(0.100 - 2x)^2} = 7.3 \times 10^{-18}$$

This is a cubic equation (one term involves x^3) and can be rather difficult to solve unless we can simplify it. In this instance we are able to do so because the very small value of K_c tells us that hardly any of the H_2O will decompose. Whatever the value of x, we know that it is going to be very small. This means that 2x will also be small, so when this tiny value is subtracted from 0.1, the result will still be very very close to 0.1. We will make the assumption, then, that the denominator be essentially unchanged from 0.100 by subtracting 2x from it; that is, we will assume that $0.1 - 2x \approx 0.1$.

This assumption greatly simplifies the math. We now have

$$\frac{4x^3}{(0.100)^2} = 7.3 \times 10^{-18}$$

$$4x^3 = (0.0100)(7.3 \times 10^{-18}) = 7.3 \times 10^{-20}$$

$$x^3 = 1.8 \times 10^{-20}$$

$$x = \sqrt[3]{1.8 \times 10^{-20}}$$

$$= 2.6 \times 10^{-7}$$

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Notice that : the value of x that we've obtained is indeed very small. If we and subtract the answer from 0.100, we still get 0.100 when we round to the correct number of significant figures (the third decimal place).

$$0.1 - 2x = 0.1 - 2(2.6 \times 10^{-7}) = 0.09999948$$

$$= 0.1 \text{ (rounded correctly)}$$

This check verifies that our assumption was valid. Finally, we have to obtain the concentration. Our table gives: $[H_2] = 2x$

Therefore, $[H_2] = 2(2.6 \times 10^{-7}) = 5.2 \times 10^{-7}$ M

The simplifying assumption made in the preceding example is valid because a very small number is subtracted from a much larger one. We could also have neglected x (or 2x) if it were a very small number that was being added to a much larger one. Remember that you can only neglect an x that's added or subtracted; you can never drop an x that occurs as a multiplying or dividing factor. Some examples are:

As a rule of thumb, you can expect that these simplifying assumptions will be valid if the concentration from which x is subtracted, or to which xis added, is at least 1000 times greater than K. For instance, in the preceding example, 2x was subtracted from 0.100. Since 0.100 is much larger than 1000 x (7.3×10^{18}) we expect the assumption $0.1 - 2x \approx 0.1$ to be valid. However, even though the simplifying assumption is expected to be valid, always check to see if it really is after finishing the calculation. If the assumption proves invalid, then some other way to solve the algebra must be found.

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Exercises

1. Write the equilibrium law for each of the following reactions in terms of molar concentrations.

(a) $H_2(g) + Cl_2(g) = 2HCl(g)$

(b) $\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) = HCl(g)$

How does K_c for reaction (a) compare with K_c for reaction (b)?

2. The reaction $CO(g) + 2H_2(g) = CH_3OH(g)$ has $K_p = 6.3 \times 10^{-3}$ at 225 °C. What is the value of K_c at this temperature?

3. The reaction $N_2O(g) + NO_2(g) = 3NO(g)$ has $K_c = 4.2 \times 10^{-4}$ at 500 °C. What is the value of K_p at this temperature?

4. The heterogeneous reaction $2HCl(g) + I_2(s) = 2HI(g) + Cl_2(g)$ has $K_c = 1.6 \times 10^{-34}$ at 25 °C. Suppose 0.1 mol of HCl and solid I_2 is placed in a 1L container. What will be the equilibrium concentrations of HI and Cl_2 in the container?

5. How will the equilibrium:

$$\text{heat} + CH_4(g) + 2H_2S(g) = CS_2(g) + 4H_2(g)$$

be affected by the following?

(a) The addition of $CH_4(g)$ **(b) The addition of $H_2(g)$**

(c) The removal of $CS_2(g)$ **(d) A decrease in the volume of the container**

(e) An increase in temperature

6. Consider the equilibrium

$$N_2O(g) + NO_2(g) = 3NO(g) \qquad \Delta H° = +155.7\ kJ$$

In which direction will this equilibrium be shifted by the following changes?

(a) Addition of N_2O (c) Addition of NO

(b) Removal of NO_2

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(d) Increasing the temperature of the reaction mixture

(e) Addition of helium gas to the reaction mixture at constant volume

(f) Decreasing the volume of the container at constant temperature

7. At a certain temperature, $K_c = 0.18$ for the equilibrium

$PC1_3(g) + Cl_2(g) = PCl_5(g)$ Suppose a reaction vessel at this temperature contained these three gases at the following concentrations:

$[PC1_3] = 0.042$ M $[Cl_2] = 0.0240$ M, $[PC1_5] = 0.005$ M.

a) Is the system in a state of equilibrium?

b) If not, which direction will the reaction have to proceed to get to equilib.?

8. The reaction: $N_2O_4(g) = 2NO_2(g)$ has $K_p = 0.14$ 25 °C

In a reaction vessel containing these gases in equilibrium at this temperature, the partial pressure of N_2O_4 was 0.250 atm.

(a) What was the partial pressure of the NO_2 in the reaction mixture?

(b) What was the total pressure of the mixture of gases?

9. The reaction: $2HC1(g)$ $H_2(g) + Cl_2(g)$ has $K_c = 3.2 \times 10^{-34}$ at 25 C.

If a reaction vessel contains initially 0.0500 mol/L of HC1 and then reacts to reach equilibrium, what will be the concentrations of H_2 and Cl_2?

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Answers for Exercises

1 (a) $K_c = \frac{[HCl]^2}{[H_2][Cl_2]}$ (b) $K_c = \frac{[HCl]}{[H_2]^{1/2}[Cl_2]^{1/2}}$

K_c for reaction (b) is the square root of K_c for action (a).

2 $K_c = 11$

3 $K_p = 2.7 \times 10^{-2}$

4) $[HI] = 1.47 \times 10^{-12}$ M , $[Cl_2] = 7.37 \times 10^{-13}$ M

5 a) The system shifts to the right to cosume some of the added methane.

b) The system shifts to the left to cosume some of the added hydrogen.

c) The system shifts to the right to make some more carbon disulfide.

d) The system shifts to the left to decrease the amount of gas.

e) The system shifts to the right to absorb some of the added heat.

6 a) right b) left c) left d) right

e) no effect f) left

7) a)This is not the value of the equilibrium constant, and we conclude that the system is not as. Equilibrium.

b) Since the value of the reaction quotient for this system is larger than of the equilibrium constant, the system must shift to the left to reach equilibrium.

8) a) $P_{NO2} = 0.187$ atm b) $P_{total} = 0.437$ atm

9) $[H_2] = [Cl_2] = 8.9 \times 10^{-19}$ M

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